Europass Curriculum Vitae

Personal information



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Work experience

Sep 2024 – Apr 2026

Associate Robotics Engineer

Strider Robotics, Bengaluru, India

- Migrated a quadruped's legacy ROS1 state estimator to ROS2 to support the robot's updated software stack, validating the Kalman Filter's output against motion-capture ground truth to confirm estimation accuracy.
- Ported the robot's simulation from the end-of-life Gazebo Classic to Gazebo Harmonic and migrated internal custom plugins from gz-classic to the gz-sim framework, extending full simulation support.
- Developed an internal ROS2 SDK and standardized robot software configurations, eliminating manual setup steps and significantly reducing bring-up time for control and navigation experiments.
- Estimated motor and gearbox parameters using CMA-ES and updated Isaac Lab's actuator model to match the real quadruped, narrowing the sim-to-real gap for RL policy training.
- Trained an actuator-net on real hardware data to model motor torque nonlinearities; together with the calibrated simulation, reduced thermal load by 25% and eliminated overheating-related hardware failures during slope walking.
- Designed and implemented push- and fall-recovery behaviors for a quadrupedal robot, enabling it to withstand slips and external pushes and to self-right from a fully inverted (upside-down) position.

Jan 2024 – Aug 2024

Robotics Intern

Strider Robotics, Bengaluru, India

- Integrated the Unitree SDK for the GO1 and GO2 quadruped with robot agnostic ROS2 stack, enabling rapid hardware deployment and testing.
- Developed actuator-specific ros2_control hardware interfaces and agnostic controllers to test multiple QDD actuators on a common setup, benchmarking speed, torque, and efficiency to inform actuator selection.

- Implemented a model-based computed torque controller for quadruped leg trajectory tracking, reducing mean tracking error by 10% relative to a PD controller tracking baseline.

Education and training

Sep 2026 – Present

M.Sc. in Engineering Cybernetics

Universität Stuttgart, Stuttgart, Germany

Jun 2024

B.E. in Electronics and Communication (CGPA: 2.1/5.0)

Gujarat Technological University, India

Thesis: *Computed torque control based trajectory tracking for legged locomotion*

Leadership and activities

Aug 2022 – Nov 2023

Software Lead

GTU Robotics Club (GRC 2023), Ahmedabad, India

- Led a software team to build a modular stack with omnidirectional navigation and dead-reckoning localization, winning DD Robocon 2023 (1st of 63) and representing India at ABU Robocon 2023 (6th of 13).

Open source contributions

Maintainer

RViz Dynamic Reconfigure

- Developed an RViz2 plugin for live GUI-based reconfiguration and logging of ROS2 node parameters, streamlining tuning and debugging.

Personal skills

Programming languages

C++, Python, Bash

Software / tools

ROS, ROS 2, CMake, Git, Gazebo, MuJoCo, NVIDIA Isaac Lab, MoveIt2, RViz2, Nav2

Certified (IELTS) European level[(*)]

English

Understanding		Speaking		Writing
Listening	Reading	Spoken interaction	Spoken production	
C2 Proficient user	C2 Proficient user	C2 Proficient user	C2 Proficient user	C2 Proficient user

[(*)] *Common European Framework of Reference (CEF) level, as equivalent to IELTS result*